

# JavaScript Operators

## What is an Operator?

An **Operator** is a symbol that performs an operation on one or more values (operands).

```
let sum = 10 + 5;  
  
console.log(sum); // 15
```

## Types of Operators

1. Arithmetic Operators
2. Assignment Operators
3. Comparison Operators
4. Logical Operators
5. Increment & Decrement Operators
6. String Operator
7. Ternary Operator
8. Type Operator

## 1. Arithmetic Operators

Used for mathematical calculations.

| Operator | Meaning        | Example |
|----------|----------------|---------|
| +        | Addition       | 10 + 5  |
| -        | Subtraction    | 10 - 5  |
| *        | Multiplication | 10 * 5  |
| /        | Division       | 10 / 5  |

| Operator | Meaning  | Example |
|----------|----------|---------|
| %        | Modulus  | 10 % 3  |
| **       | Exponent | 2 ** 3  |

```

let a = 10;
let b = 3;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
console.log(a % b);
console.log(a ** b);

```

## 2. Assignment Operators

Used to assign or update values.

| Operator | Example | Same As      |
|----------|---------|--------------|
| =        | x = 5   | Assign value |
| +=       | x += 2  | x = x + 2    |
| -=       | x -= 2  | x = x - 2    |
| *=       | x *= 2  | x = x * 2    |
| /=       | x /= 2  | x = x / 2    |
| %=       | x %= 2  | x = x % 2    |

```
let x = 10;

x += 5;
console.log(x); // 15

x -= 3;
console.log(x); // 12
```

### 3. Comparison Operators

Used to compare two values.

| Operator           | Meaning               | Example                 |
|--------------------|-----------------------|-------------------------|
| <code>==</code>    | Equal (value only)    | <code>5 == "5"</code>   |
| <code>===</code>   | Strict Equal          | <code>5 === "5"</code>  |
| <code>!=</code>    | Not Equal             | <code>5 != 4</code>     |
| <code>!==</code>   | Strict Not Equal      | <code>5 !== "5"</code>  |
| <code>&gt;</code>  | Greater Than          | <code>10 &gt; 5</code>  |
| <code>&lt;</code>  | Less Than             | <code>5 &lt; 10</code>  |
| <code>&gt;=</code> | Greater Than or Equal | <code>5 &gt;= 5</code>  |
| <code>&lt;=</code> | Less Than or Equal    | <code>5 &lt;= 10</code> |

```
console.log(10 > 5);
console.log(10 == "10");
console.log(10 === "10");
```

### 4. Logical Operators

Used to combine conditions.

| Operator | Meaning |
|----------|---------|
| &&       | AND     |
| ,        |         |
| !        | NOT     |

```
let age = 20;
let hasID = true;

console.log(age >= 18 && hasID);
console.log(age < 18 || hasID);
console.log(!hasID);
```

## 5. Increment & Decrement Operators

Increase or decrease a value by 1.

| Operator | Meaning   |
|----------|-----------|
| ++       | Increment |
| --       | Decrement |

```
let count = 5;

count++;
console.log(count); // 6

count--;
console.log(count); // 5
```

## 6. String Operator

The `+` operator joins strings.

```
let firstName = "MD";  
let lastName = "Rokonuzzaman";  
  
console.log(firstName + " " + lastName);
```

Output:

```
MD Rokonuzzaman
```

## 7. Ternary Operator

A short way to write an `if...else` .

### Syntax

```
condition ? trueValue : falseValue;
```

```
let age = 20;  
  
let result = age >= 18 ? "Adult" : "Minor";  
  
console.log(result);
```

## 8. Type Operator

The `typeof` operator returns the data type of a value.

```
console.log(typeof "Hello");  
console.log(typeof 100);  
console.log(typeof true);
```

```
console.log(typeof []);  
console.log(typeof {});
```

## Operator Precedence

Operators with higher precedence execute first.

```
let result = 10 + 5 * 2;  
  
console.log(result);
```

Output:

```
20
```

Because `*` has higher precedence than `+`.

## Summary

| Category            | Purpose                      |
|---------------------|------------------------------|
| Arithmetic          | Mathematical calculations    |
| Assignment          | Assign or update values      |
| Comparison          | Compare values               |
| Logical             | Combine conditions           |
| Increment/Decrement | Increase or decrease by 1    |
| String              | Join strings                 |
| Ternary             | Short <code>if...else</code> |
| Type                | Check data type              |

## Easy Way to Remember

- `+ - * / % **` → Arithmetic
- `= += -= *= /= %=` → Assignment
- `== === != > < >= <=` → Comparison
- `&& || !` → Logical
- `++ --` → Increment / Decrement
- `+` → String Concatenation
- `? :` → Ternary
- `typeof` → Type Checking

## Interview Questions

### Q1. What is the difference between `==` and `===`?

- `==` compares **only values** (type conversion occurs).
- `===` compares **both value and data type**.

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

### Q2. What is the difference between `&&` and `||`?

- `&&` (AND): **All conditions must be true.**
- `||` (OR): **At least one condition must be true.**

```
console.log(true && false); // false
console.log(true || false); // true
```